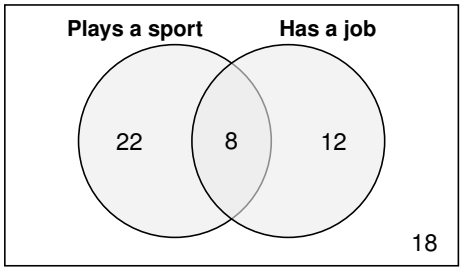
Two-Way Tables and Relative Frequency KEY

Name _____

60 students answered two questions: *Do you play a sport? Do you have a job?* A **two-way table** shows the answers to both questions at once.

	Job	No job	Total
Sport	8	22	30
No sport	12	18	30
Total	20	40	60



Each **cell** counts students who match **both** labels. 8 students play a sport **and** have a job.
The **totals** are in the margins. 30 students play a sport. 20 have a job.
The **grand total** is the corner. 60 students in all.

Check: Each row adds to its total. Each column adds to its total. $8 + 22 = 30$, $8 + 12 = 20$.

Venn diagram: the overlap is "both" (8). Outside the circles is "neither" (18).

Relative frequency table. Divide every count by the **grand total**. This gives the percent of **all** students.

	Job	No job	Total
Sport	$\frac{8}{60} \approx 13\%$	$\frac{22}{60} \approx 37\%$	$\frac{30}{60} = 50\%$
No sport	$\frac{12}{60} = 20\%$	$\frac{18}{60} = 30\%$	$\frac{30}{60} = 50\%$
Total	$\frac{20}{60} \approx 33\%$	$\frac{40}{60} \approx 67\%$	100%

About 13% of **all** the students play a sport and have a job. The four inside cells add to 100%.

Work with a partner. Compare answers when you finish each problem.

1. 80 students said how they get to school.

	Bus	Car	Walk	Total
9th grade	18	12	10	40
10th grade	14	20	6	40
Total	32	32	16	80

   (a) How many 10th graders walk? 6

   (b) How many students ride the bus? 32

   (c) How many students are 9th graders who come by car? 12

   (d) How many students do **not** walk? $80 - 16 = 64$

2. Fill in the missing numbers. Students said if they ate breakfast today.

	Yes	No	Total
11th grade	15	20	35
12th grade	12	10	22
Total	27	30	57

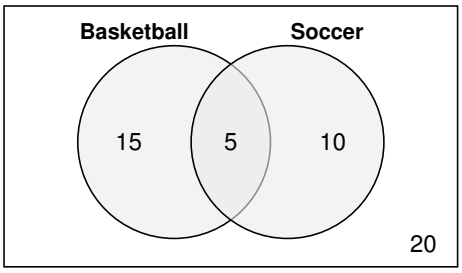
3. 50 people were asked about pets. 30 like cats. 26 like dogs. 12 like both.

   (a) Fill in the two-way table.

	Likes dogs	No dogs	Total
Likes cats	12	18	30
No cats	14	6	20
Total	26	24	50

   (b) How many people like neither cats nor dogs? 6

4. This Venn diagram shows which sports 50 students play.



Make a two-way table from the Venn diagram.

	Soccer	**No soccer**	**Total**
Basketball	5	15	20
No basketball	10	20	30
Total	15	35	50

5. Make a relative frequency table for problem 1. Divide each count by 80. Write percents.

	Bus	**Car**	**Walk**	**Total**
9th grade	22.5%	15%	12.5%	50%
10th grade	17.5%	25%	7.5%	50%
Total	40%	40%	20%	100%

(a) What percent of all the students are 10th graders who come by car? 25%

(b) What percent of all the students walk? 20%